Transformer Accounting: Simplifying Parameter and Compute Analysis

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1 Introduction

There are some insights that simplify thinking about the transformer quite a bit.

2 Parameter Conventions

The following conventions are used throughout this analysis:

$$d_{ff} = 4 \cdot d_{model} \tag{1}$$

$$d_q = d_k = d_v = \frac{d_{model}}{n_{heads}} \tag{2}$$

We consider a non-gated MLP architecture.

3 Parameter Structure

3.1 Matrix Organization

There are 6 matrices with parameters per layer. They all have similar shapes:

- **Two for MLP:**
  - First matrix: $d_{model} \to d_{ff}$
  - Second matrix: $d_{ff} \to d_{model}$
- **Four for attention:**
  - All have shape $d_{model} \to d_{model}$

3.2 Total Parameters

The total number of parameters per layer is:

$$\text{Parameters} = 2 \cdot (d_{model} \cdot d_{ff}) + 4 \cdot d_{model}^2 \tag{3}$$

For the common case where $d_{ff} = 4 \cdot d_{model}$:

$$\text{Parameters} = 12 \cdot d_{model}^2 \tag{4}$$

In a sufficiently large model, this formula will get you within 1% of non-embedding parameters.

4 Computational Analysis

4.1 Matrix Multiplication Pattern

Each parameter matrix is multiplied by each token embedding on each pass. The usage pattern is remarkably simple: all 6 matrices multiply the same inputs.

The input to a model has shape $[\text{batch}, \text{sequence}, d_{model}]$. It consists of d_{model} vectors, where each vector is multiplied by each parameter matrix.

4.2 Compute Requirements

The total compute for multiplying with all matrices is:

$$\text{Compute} = 24 \cdot d_{model}^3 \cdot \text{seq} \cdot \text{batch} \tag{5}$$

The compute per token is proportional to $2 \times \text{num\_params}$. Arithmetic intensity is directly linear to the number of tokens processed.

5 Attention Mechanism

There is additional compute required to handle interactions across the sequence, known as the attention mechanism.

Key properties of attention:

- Does **not** add any parameters
- Compute is linear in d_{model}
- Compute is quadratic in sequence length

6 Parameter and Compute Summary

Operation	**Parameters**	**FLOPs per Token**
MLP Layer 1	$d_{model} \times d_{ff}$	$2d_{model} \times d_{ff}$
MLP Layer 2	$d_{ff} \times d_{model}$	$2d_{ff} \times d_{model}$
Attention: Q	$d_{model} \times d_{model}$	$2d_{model}^2$
Attention: K	$d_{model} \times d_{model}$	$2d_{model}^2$
Attention: V	$d_{model} \times d_{model}$	$2d_{model}^2$
Attention: Output	$d_{model} \times d_{model}$	$2d_{model}^2$
Attention: Scores	—	$2d_{model} \times \text{seq}^2$
Total per Layer	$\mathbf{2d_{model} \times d_{ff} + 4d_{model}^2}$	$\mathbf{2d_{model}(d_{ff} + 2d_{model}) + 2d_{model} \times seq^2}$
Common Case ($d_{ff} = 4d_{model}$)	$\mathbf{12d_{model}^2}$	$\mathbf{24d_{model}^2 + 2d_{model} \times seq^2}$

Table 1: Parameter count and compute requirements per transformer layer